Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

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***Abstract*—Although airfoils are one of the most fundamental components of aerospace engineering, relatively little is known about their aerodynamic interactions in biplane configurations, as most studies tend to solely focus on monoplanes. This study addresses this gap by analyzing airfoils in both mono and biplane configurations. The study focuses on four different airfoils: NACA 0012, 2212, 2412, and 2612. For initial theoretical analysis, 2D CFD (computational fluid dynamic) simulations of these airfoils were conducted in XFLR5 and ANSYS Fluent Flow. Following the computational study, 3D-printed models were tested in a wind tunnel at a Reynolds number of 20,100 for experimental analysis. 51 biplane configurations were tested in combinations determined by airfoil shape and decalage test matrices. Furthermore, finite-span aspect-ratio corrections were applied to the CFD results to distinguish 3D downwash effects from the behavior of low-Reynolds-number viscous flow. The experimental results demonstrated that aerodynamic efficiency for airfoil shape optimization is influenced more through changes in lift rather than in drag. The best performing airfoil shape combination was further optimized through the decalage test matrix. Finally, the highest performing configuration was determined to be NACA 2212 on the top at 0-degree angle of attack and NACA 2212 on the bottom at 5-degree angle of attack. The most efficient combinations generally used a converging wing combination that accelerated airflow through the gap, creating a concentrated Venturi effect, while negative lower-wing angles caused noticeable losses in lift and an overall worse performance.**

***Index Terms*—Airfoil simulations, Airfoils, Biplanes, Computational fluid dynamics, Flow control, Flow interference, Low Reynolds number, Wind tunnel testing**

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I. Introduction

The idea of biplanes (two wings stacked one above the other) has been considered and studied ever since the inception of airplanes. The main motivation behind early biplanes was to reduce structural weight by bracing two shorter wings against one another rather than cantilevering a single long wing [1], [2]. Over time they lost popularity in favor of monoplanes (one-winged airplanes).

The motivation to focus on biplanes, rather than monoplanes, stemmed from the limited modern research in the field. Biplanes can be useful alternatives to monoplanes in many applications due to their distinct behavior and structure, such as for crop dusting and landing on short airfields, partly due to their greater maneuverability, shorter wingspans, and higher lift at low speeds. A computational and experimental analysis and optimization of factors, such as stagger, decalage, gap, and airfoil shape, has the potential to reveal new insights significant to the field of aerospace engineering as a whole. The novelty is not just a source of the possible applications, but a supplement for the lack of understanding in biplane aerodynamics which can provide deeper understanding of countless physical phenomena and aspects crucial to aerospace engineering [1].

II. Background

Each wing can first be modeled as a line segment carrying circulation, a simplification that later generalizes to real airfoil shapes. Without circulation, the flow around a cambered section would turn the sharp trailing edge at infinite velocity, which is not physical. Superposing a "circulation flow" introduces an equal and opposite singularity there, so the two cancel and the flow leaves the trailing edge smoothly. For a biplane this condition must hold at both trailing edges

simultaneously, which is accomplished by mapping the two wings into an equivalent "tandem" arrangement and solving for the circulation strengths that satisfy it. This yields the biplane lift:

$$C_L = 2\pi \sin\beta\, B + 2\pi \sin\beta_0 \sqrt{B} \tag{1}$$

The angle of attack for a biplane is denoted by β, while β_0 is the effective angle representing the airfoil's camber. Additionally, B is the factor that represents the loss of lift in biplanes (which is always less than 1). The first term is representative of lift driven by the sectional angle of attack α, the angle between a wing's chord line and the oncoming relative wind, and the second term is representative of camber-driven lift [1].

For upper and lower wings with different amounts of camber, each wing has its own effective-curvature angle, but the total lift given by the equation above only depends on the average of the two. The difference between the two wings' incidence[1] angles is the decalage of the biplane; because thin-airfoil theory treats effective curvature as an additive contribution to incidence, a difference in camber between the wings acts in the same way. For small decalage, the average of the two wings' α can be used to determine the coefficient of lift, so in theory decalage does not significantly influence total lift. However, the simplifications made in the two-dimensional treatment above cannot describe drag contributions that ultimately govern the aerodynamic efficiency of the biplane.

To understand the effective contributions of decalage to the overall drag of the system, two distinct mechanisms must be considered. The first is sectional interference, in which the pressure fields of the upper and lower wings overlap across the gap[2], altering the loading on each section and the flow through the channel between them. This type of interference is mostly governed by the physical gap distance between each wing, the decalage, and the sectional airfoil geometry. The second type of interference is induced-drag interference. Induced drag is the product of downwash, the downward velocity induced by a finite wing's trailing vortices, which tilts the local relative wind and rotates the lift vector rearward [3]. In a biplane, the trailing vortex systems of the two wings interact in the far field, reducing the net downwash and therefore the induced drag relative to a single wing of the same span producing the same total lift [1], [4]. This mechanism is a finite-span effect and vanishes in two dimensions where the wings are assumed to be of infinite span.

III. Experimental Procedure

A. Simulations

Each individual airfoil tested was first simulated in XFLR5 and ANSYS. 16 airfoils were first screened in XFLR5, which couples an inviscid linear-vorticity panel method to a two-equation integral boundary layer formulation through a viscous-inviscid interaction scheme, following XFOIL [5]. This produced the drag polar and the sectional lift curve for each airfoil. The six highest performing airfoils in XFLR5 were simulated in ANSYS Fluent Flow, which solves the Reynolds-averaged Navier-Stokes equations with a turbulence closure and additionally generates streamline and velocity contour graphics, along with lift and drag coefficient calculations at various angles of attack.

Both tools were run in two-dimensional, sectional mode in this study. Neither configuration therefore contains downwash or any other finite-span effect which is the reason the finite-span treatment of Sec. III-B is required before comparison with three-dimensional behavior.

B. Aspect Ratio / Finite Span Correction

Because the ANSYS simulations were two-dimensional while the wind-tunnel models had a finite span, a finite-span correction was applied to the simulated results before they were compared with the experimental data. This helped separate finite-span effects from other experimental differences. The rectangular planform models had a span (b) of 150 mm and a chord (c) of 30 mm, giving an aspect ratio of $AR = b/c = 5$. For each airfoil, the 2D lift-curve slope, a_0, was obtained from the linear pre-stall portion of its ANSYS C_l vs. α data. The ANSYS slope was used instead of the ideal value of 2π, because it reflected the low-Reynolds-number conditions of the study. The corrected slope was calculated from lifting-line theory [3], [6] as

$$a_{3D} = \frac{a_0}{1 + \frac{a_0}{\pi AR e}}, \tag{2}$$

using an assumed Oswald efficiency factor of $e = 0.8$ for an unswept rectangular wing. The uncertainty associated with this assumption is discussed in the limitations section. Predicted losses from the pre-stall slopes ranged from 13.1% for NACA 2612 to 22.0% for NACA 2212. Using the ANSYS zero-lift angle α_{L0}, the corrected lift coefficient was approximated by

$$C_{L,3D}(\alpha) = a_{3D}(\alpha - \alpha_{L0}). \tag{3}$$

Angles were converted to radians when the lift-curve slopes were measured in rad^{-1}.

Drag was estimated by adding the induced contribution to the sectional value. For a multiplane this contribution is scaled by Munk's interference factor k_M, the ratio of the biplane induced drag to that of a reference monoplane of the same span carrying the same total lift [1], [4], giving

$$C_{D,3D} = C_{D,2D} + k_M \frac{C_{L,3D}^2}{\pi AR e}. \tag{4}$$

Setting $k_M = 1$ recovers the monoplane result, so (4) applies to both configurations.

The corrected values were compared with the wind-tunnel data. Remaining differences could therefore be examined in relation to turbulence, surface roughness, mounting interference, and wingtip leakage near the tunnel walls. The correction was only applied in the linear pre-stall region and was not used to predict stall.

[1]The tilt angle of each wing compared to the fuselage center line

[2]The vertical distance between the upper and lower wings

C. Wind Tunnel Procedure

Using CAD software from Onshape, 4 different types of airfoils (NACA: 0012, 2212, 2412, and 2612) were imported, modified to include a small hole to insert a probe at six different α, modeled, and 3D printed. Angles of attack were manually calibrated using a precision protractor before testing to ensure that the alignment was accurate across all trials [7].

For the biplane configuration, a bracket was created to hold the two airfoils at a gap of 30 mm with no stagger. The hole for the probe was placed at the quarter-chord point in the center of the wing span, which approximates the aerodynamic center of the airfoil [8]. The reported biplane forces are those of the combined two-wing assembly, referenced to the planform area of a single wing.

For wind tunnel experimentation, a test matrix was created that tested every possible combination of airfoil shapes at $0°$ angle of attack. From the most efficient airfoil combination, a second test matrix was created to evaluate these combinations at varying angles of attack to determine the best angle of attack and the critical angle of attack for each airfoil combination. Each airfoil configuration in each test matrix was placed in the wind tunnel. The chord-based Reynolds number was $Re = U_\infty c/\nu = 20{,}100$, corresponding to a freestream velocity of approximately $10\,\text{m/s}$ at the $30\,\text{mm}$ chord.

The models spanned the test section to within $0.5\,\text{mm}$ at each tip, which suppresses the formation of a trailing vortex system and the downwash associated with it. The measured forces therefore approach two-dimensional sectional values, and the tip clearance admits a leakage jet rather than a rolled-up vortex sheet. This condition is the opposite of the one treated by Moreno et al. [9], who evolve the trailing vorticity of a biplane as a pair of coupled vortex sheets in an unbounded Trefftz plane and extract the induced-drag interference factor k_M from the kinetic energy of the wake. That analysis requires a wake that is free to roll up, which the present configuration deliberately removes. The induced-drag interference it describes is therefore absent from the measurements reported here, and differences observed between biplane configurations are attributable to the sectional mechanism alone. Accordingly, k_M in (4) should be read as a free-span prediction applied to the corrected simulation data, and not as a quantity that these wind-tunnel results either confirm or contradict.

IV. Results and Discussion

A. Monoplane Results

Fig. 1 shows the experimental lift curves for the four monoplane airfoils, obtained by fitting a polynomial to the measured C_L at each angle of attack. All four exhibit the expected qualitative behavior: an approximately linear region at low α, followed by a reduction in lift-curve slope as the flow approaches stall. Quantitatively, however, the measured values depart from the 2D simulations and from classical predictions, most notably in the zero-lift angles of the cambered sections. Several contributing factors are considered below, though no single cause is isolated for the zero-lift offsets.

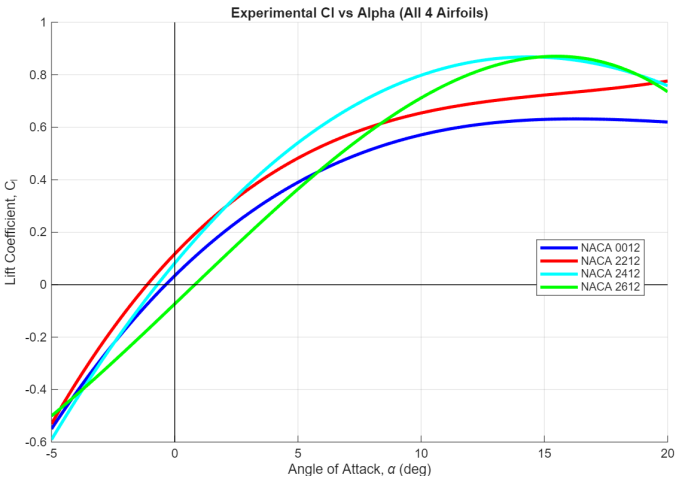


Fig. 1. C_L vs. α for all 4 airfoils overlaid

Compared to each other, NACA 2212 dominates at positive angles of attack, but NACA 2612 has the highest overall C_L and tends to dominate after $15°$. Despite all of the airfoils having significantly different values than the simulations, the rate at which the C_D or C_L change, as shown in Fig. 2, with respect to α remained very similar, showing that the general trend remained for all the airfoils except NACA 2212.

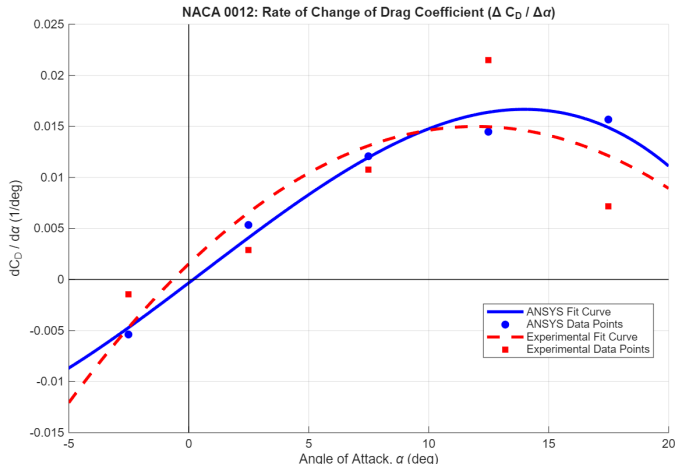


Fig. 2. Rate of Change of Drag Coefficient ($\Delta C_D/\Delta\alpha$)

The NACA 2212 airfoil did not stall as predicted in the simulations at $10°$. This occurred due to the formation of short laminar separation bubbles on the surface of the airfoil, delaying the stall beyond the range tested and increasing the lift [10].

For all 4 NACA airfoil types tested, individual graphs of C_L vs. α were also created, with the simulation and experimental results plotted, and velocity contour and streamline diagrams were created by the simulations. The measured lift exceeded the simulated values for α between $0°$ and $15°$, and fell below them between $-5°$ and $0°$. Two mechanisms are consistent with these measurements. First, solid and wake blockages in a wall-bounded test section raise the effective dynamic

pressure the wing models experience and constrains curvature of the streamlined flow, both of which increase the measured lift relative to an unconfined test section [7]. Second, at this Reynolds number the integral and RANS closures used here reproduce laminar separation bubbles poorly, and a bubble that reattaches raises lift above the simulated value [10], [11].

B. Biplane Wind Tunnel Results

For all four NACA airfoils selected, a test of every possible combination was done: 16 different combinations. The results are demonstrated through the bar chart in Fig. 3. The notation utilized to describe a biplane configuration will be [Top Airfoil 4-digit NACA code] $\times$ [Bottom Airfoil 4-digit NACA code]; i.e., 0012 $\times$ 0012. The combination with the highest lift-to-drag ratio is 2212 $\times$ 2212. The lift-to-drag ratio for the combination was 1.11.

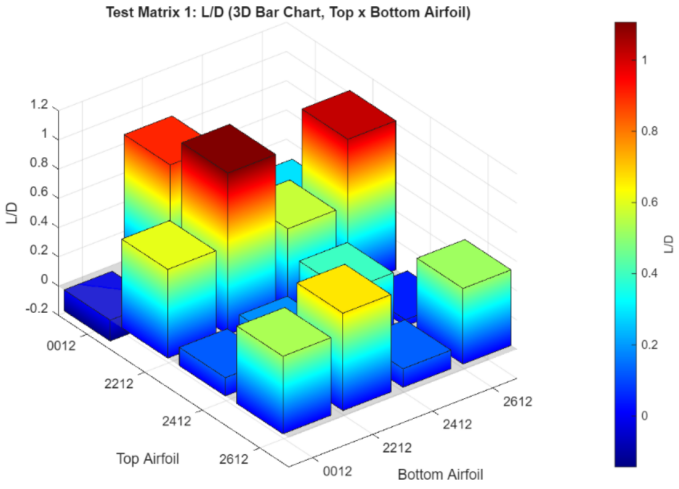


Fig. 3. Test matrix 1: L/D (3D Bar Chart, Top $\times$ Bottom Airfoil)

All combinations of airfoils, except the 2 most efficient, produced more drag than lift, with 2212 $\times$ 2212 determined as the best configuration. The worst performing combination is 0012 $\times$ 0012. As 0012 is the symmetric airfoil, a zero angle of attack does not result in any lift, so all that is measured is drag and interference.

The results of the second matrix are demonstrated in the bar chart in Fig. 4. 36 different combinations were created by varying decalage for the top and bottom airfoils of the 2212 $\times$ 2212 combination. From this, the best lift-to-drag ratio was found. This was done with the top airfoil at 0° and the bottom at 5° which resulted in a ratio of 9.12. The other well performing combinations were $5^\circ \times 10^\circ$ (L/D = 8.98) as well as $5^\circ \times 5^\circ$ (L/D = 6.19).

Fig. 5 reveals that the efficiency ratio is tracked by C_L rather than C_D, as illustrated by the green line following the same trend as the bars C_L. The 2212 $\times$ 2212 combination can optimize camber without causing too much airflow interference between the top and bottom wings. A higher max-camber location combination may have increased lift, but there would be greater interference from the bottom wing, which would cause an overall reduction in lift. Equation (1) is consistent

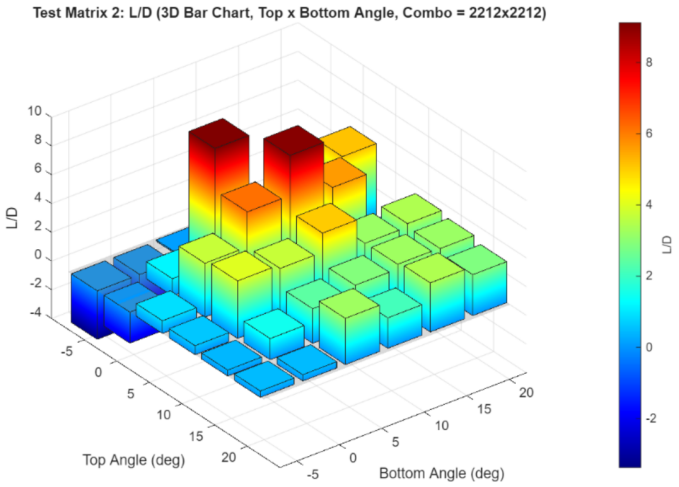


Fig. 4. Test matrix 2: L/D (3D Bar Chart, Top $\times$ Bottom Angle, 2212 $\times$ 2212)

with the observed ordering to the extent that it depends on the average effective camber of the pairing, though it depends only on that average and therefore does not by itself discriminate between pairings of equal mean camber.

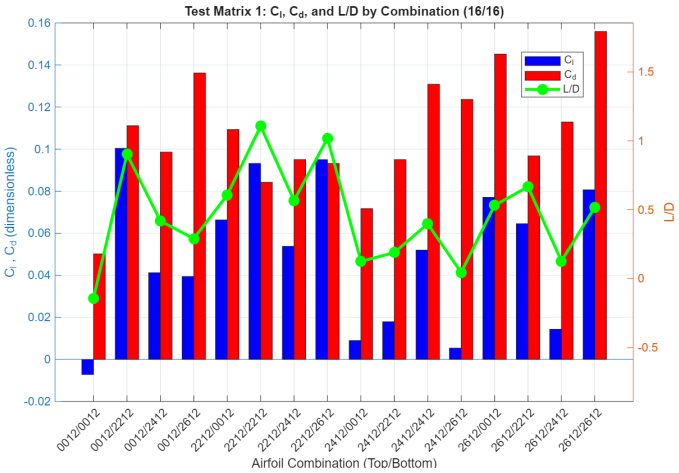


Fig. 5. C_L (blue) and C_D (red) tracked against aerodynamic efficiency (green)

The three worst-performing decalage combinations did not particularly produce a large amount of drag; rather, the inefficiency came from the reduction in lift force due to earlier flow separation from negative α. The monotonic increase of lift with mean incidence over the pre-stall range is consistent with (1), which depends on the average incidence of the two wings; the equation does not predict drag and therefore cannot account for the efficiency ordering, which is set by the accompanying growth in drag.

1) Further Discussion on 2212 $\times$ 2212 at $0^\circ \times 5^\circ$ and $5^\circ \times 10^\circ$: The biplanes at $0^\circ \times 5^\circ$ and $5^\circ \times 10^\circ$ performed significantly better than the other combinations of angles. They had an L/D ratio of 9.12 and 8.98, respectively. These

biplanes both shared a converging geometry on the channel between the airfoils. The local acceleration of the air through the converging channel is expected to reshape the pressure distribution on the wing surfaces. This creates a more concentrated Venturi effect. No surface-pressure measurements were taken, so this mechanism is inferred from the force data and the gap geometry rather than observed directly; resolving it would require a pressure-tapped model or particle image velocimetry of the channel.

2) Stall-Order Asymmetry: The data from the second test matrix suggest an asymmetrical stall relationship between the two airfoils. All of the best performing biplanes had a greater angle of attack on the bottom airfoil relative to the top, while all of the worst performing had the greater angle on top. Stalling the top airfoil therefore costs more lift than stalling the bottom one. Because the two wings reach stall at different incidences, the lower wing continues to produce lift as the upper one stalls, which can help return an aircraft to a stable angle of attack.

3) Monoplane and Biplane Configuration Performance Comparison: For the specific test conditions of the experiments, the optimal biplane (2212×2212 at $0^\circ \times 5^\circ$) performed better by a significant margin compared to the optimal monoplane (2612 at 10°). The L/D was 9.12 and 4.21, respectively, with a difference of 4.91. While promising, more robust testing and consideration of the test-conditions are necessary for certain applications.

C. Limitations

Throughout testing, certain variables were assumed rather than experimentally measured. A turbulence intensity (T.I.) of 0.5% was used in ANSYS simulations instead of the wind tunnel's actual value. This affects transition location and therefore the simulated stall angle, and it is one source of the simulation-to-experiment discrepancy discussed in Sec. IV-A.

An Oswald efficiency factor ($e = 0.8$) was also assumed for finite-span corrections. Since e affects the corrected lift-curve slope and induced drag, an inaccurate value would change the correction magnitude. Wall proximity and the small wingtip gap may further alter the actual efficiency through confinement and tip-leakage effects [7].

V. Conclusions

This study evaluated the four NACA airfoils, 0012, 2212, 2412, and 2612, in monoplane and biplane configurations using computational analysis and wind-tunnel testing. The NACA 2212 $\times$ 2212 pairing proved most efficient, with the optimal decalage placing the upper wing at 0° and the lower wing at 5°. Overall, the best results paired a small positive upper-wing angle with an equal or greater lower-wing angle, demonstrating that biplane performance depends on wing-to-wing interaction rather than the behavior of isolated airfoils. Future research should continue by optimizing stagger and gap along with decalage and airfoil shape. This study's findings suggest potential applications in variable-geometry aircraft, where a second wing could improve low-speed performance before being retracted for cruise. Further optimization could be performed at supersonic speeds, eventually looking to use learned insights to implement the theoretical Busemann Biplane.

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